A Transformer-Based Roof Plane Segmentation Approach for Airborne LiDAR Point Clouds

Siyuan You, Guozheng Xu, Pengwei Zhou, Yubing Wei, Jian Yao, *Member, IEEE*, and Li Li

***Abstract*—In the fields of photogrammetry and computer vision, 3-D urban building model reconstruction from airborne light detection and ranging (LiDAR) point clouds has attracted significant attention in recent years. Accurately and automatically extracting local geometric structures, such as planar patches, from 3-D point cloud data directly determines the quality of subsequent 3-D model reconstruction. Considering that the roof is a crucial component of a real building, roof plane segmentation is a critical procedure in building 3-D reconstruction. In this article, a novel dual-branch transformer-based network is designed to accurately segment roof planes from airborne LiDAR point clouds. We first use PointNet++ followed with a transformer encoder to extract point-wise feature embeddings. Then, in the first branch, a transformer decoder module is applied to directly learn the instance centers of planar patches by giving a set of learned queries. Because the transformer can effectively model the relations of the queries and the global context information, the instance center positions of all planes included in the input point clouds can be accurately predicted. In this way, the number and center positions of roof planes are known before performing roof plane segmentation. In the second branch, we predict the offsets for each point using its point-wise feature to shift it toward the corresponding instance center. After that, the plane parameters for each plane instance can be estimated using the shifted points around the predicted centers, and the rest of the points are assigned to its nearest plane to generate the final roof planes. The experimental results illustrate that our approach can successfully address the plane segmentation challenge for diverse building roof structures while achieving performance superior to the current state-of-the-art techniques. We will make the source code of our approach publicly available at https://github.com/Li-Li-Whu/PlaneTransformer**

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Siyuan You, Guozheng Xu, Pengwei Zhou, and Yubing Wei are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: you.si.yuan@whu.edu.cn; xugzh_960508@whu.edu.cn; pengwei.zhou@whu.edu.cn; weiyubing22@gmail.com).

Jian Yao is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China, also with Nanchang Unspace Technology Company Ltd., Nanchang 330038, China, and also with Shenzhen Research Institute, Wuhan University, Shenzhen 518057, China (e-mail: jian.yao@whu.edu.cn).

Li Li is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China, and also with Shenzhen Research Institute, Wuhan University, Shenzhen 518057, China (e-mail: li.li@whu.edu.cn).

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I. Introduction

BECAUSE airborne Light Detection and Ranging (LiDAR) data can offer high-precision 3-D information of real scenes, it has been widely used for 3-D urban building reconstruction. To automatically reconstruct 3-D building models from airborne LiDAR point clouds, one of the key procedures is roof plane segmentation. The goal of plane segmentation is to accurately and automatically identify and extract the planar patches from the input point clouds, and the plane segmentation results directly determine the quality of the 3-D building models. Although the task of plane segmentation has been extensively explored in many studies [1], [2], [3], it is still a challenging problem for current 3-D building reconstruction systems [4], [5], [6] to accurately and automatically segment roof planes from point clouds.

The existing rule-based roof plane segmentation approaches can be summarized as two stages. In the first stage, they manually design some specifically handcrafted features to describe the similarities between points or patches. In the second stage, methods like Hough Transform (HT) [7], [10], RANdom SAmple Consensus (RANSAC) [11], [12], region growing [18], [19], or clustering [25], [26] are used to iteratively extract planar patches from the input point clouds based on feature similarities. In these approaches, they usually need to design some criteria to terminate the iteration of plane segmentation. Because the number and instance centers of planes included in the input point clouds are unknown before performing plane segmentation, it is difficult to set the stopping criteria and select the appropriate thresholds for the iteration in some cases. The slight change and adjustment of the predefined parameters may lead to the problems of undersegmentation and oversegmentation, especially for complex roof structures. To alleviate these problems, some approaches utilize the energy optimization-based postprocessing techniques [29], [31], [32], [33] to further improve the initial plane segmentation results. However, these approaches also need to set the convergence criteria or energy functions without the prior information of the planes to terminate or guide the optimization. In recent years, some machine learning-based roof plane segmentation approaches [40], [41], [42] have been proposed. Instead of using the handcrafted features, these approaches design the deep networks to automatically learn the features

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with higher representation ability for roof plane segmentation. After learning the features, they also use an algorithm like clustering with a certain convergence condition to extract roof planes. Therefore, the existing machine learning-based roof plane segmentation approaches still cannot provide the prior information of the planes to guide roof plane extraction. Thus, the problem of roof plane segmentation is not trivial due to the lack of prior information of planes [32].

If the prior information about the number and instance centers of planes is known before performing roof plane segmentation, we can avoid designing complex and sensitive convergence criteria to terminate the process of iteration. We can even directly assign the points to their nearest instance centers to generate the final planar patches. However, most of the plane segmentation approaches ignore this important issue. In this article, we naturally consider whether it is possible to directly determine the number and instance centers of planes from the input data before performing segmentation, thereby reducing the risks of undersegmentation and oversegmentation. Inspired by the transformer-based object detection [44], [45], [46] and instance segmentation [48], [49], [50] approaches, we propose a novel dual-branch transformer-based approach to segment roof planes from airborne LiDAR point clouds. We apply PointNet++ [52] followed by a transformer encoder to extract multiscale point-wise features. Based on this, in the first branch, a query-based transformer decoder is applied to directly predict the potential planar centers included in point clouds. In the second branch, the point-wise features are further used to predict the offset for each point to shift it toward the corresponding instance center. Then, we estimate the plane parameters using the shifted points around each predicted planar center. Finally, we allocate the unlabeled points to their nearest estimated plane using a hybrid distance-guided point assignment strategy, considering both point-to-center and point-to-plane distances, to obtain accurate roof plane segments with high geometric fidelity. Experiments on two benchmark datasets show that our approach sets a new state of the art in roof plane segmentation, with both metric scores and visual results outperforming prior works. We summarize the contributions of our work as follows.

1) Inspired by the transformers designed for object detection and instance segmentation, we propose a novel dual-branch transformer-based approach for roof plane segmentation from airborne LiDAR point clouds, which significantly outperforms the existing state-of-the-art approaches.
2) With the proposed network, we can directly detect the number and instance centers of planes from the point clouds, which are used as important prior information to guide the subsequent plane segmentation for reducing the risk of oversegmentation and undersegmentation.
3) To achieve high-quality plane segmentation results, we design a simple yet effective point assignment strategy guided by a hybrid distance metrics, jointly considering the point-to-center and point-to-plane distance, to find the optimal plane label for each point.

The remainder of this article is organized as follows. Section II reviews some existing roof plane segmentation approaches. Then, Section III introduces the details of the proposed dual-branch transformer-based approach. Section IV analyzes the ablation and comparative experimental results on two datasets. Finally, the conclusion and future prospect are provided in Section V.

II. Related Works

Recently, a number of approaches have been proposed to perform roof plane segmentation from airborne LiDAR point clouds. We divide them into two categories: rule-based and machine learning-based approaches. In addition, we briefly introduce the common transformer networks.

A. Rule-Based Approaches

The existing rule-based approaches for plane segmentation can be generally divided into four categories: model fitting-based, region growing-based, feature clustering-based, and energy optimization-based approaches.

Model fitting-based approaches aim to detect the optimal plane models from point clouds using the HT or RANSAC algorithm. HT-based approaches [7], [8], [9] first project points from Euclidean space into parametric space and then find the most probable planes by searching the local peak points in parametric space. Kurdi et al. [10] compared HT-based and RANSAC-based approaches and concluded that the latter is superior in both segmentation efficiency and quality. RANSAC-based approaches [1], [11], [12], [13], [14], [15], [16] iteratively fit the planes using random sampled points until the best planes are obtained. Li et al. [12] divided the point clouds into individual cells and analyzed the normal distribution on each cell to improve the efficiency. Yang et al. [15] introduced a novel candidate points search strategy into the RANSAC algorithm for robust plane fitting. Li et al. [16] applied RANSAC to perform rough fitting and obtain coarse plane parameters. Then, they employed principal component analysis (PCA) to eliminate the influence of outliers and generate the final fine-fitting planes. However, the above approaches still suffer from the spurious plane problem, which is caused by their inherent dependencies on the setting of stopping criteria for the iteration.

Region growing-based approaches [2], [3], [17], [18], [19], [20], [21] select the seeds in point clouds and merge their neighboring points under certain criteria to grow the planar clusters. For example, Khaloo and Lattanzi [3] performed a weighted average-based PCA in a local region for normal estimation and applied the normal vectors to guide the subsequent region growing. Gilani et al. [17] used an adapted PCA to compute accurate point normal vectors, based on which the points with sharp features, namely feature points, are detected. Then, they chose robust seeds in nonfeature points and performed region growing to group the coplanar points. Luo et al. [19] proposed to perform region growing based on supervoxel-level features extracted from point clouds to improve the efficiency. Yu et al. [21] proposed to adaptively calculate the normal vectors based on the density of point clouds and set the confidence interval between each pair of normals as the criteria of region growing. However, these approaches are quite sensitive to the selection of

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seeds and the setting of growing criteria, which may lead to issues of inaccurate segmentation of the boundaries, and undersegmentation or oversegmentation.

Feature clustering-based approaches [22], [23], [24], [25], [26], [27], [28] generally use a clustering method to extract roof planes from point clouds with manually designed features. He et al. [23] adopted spectral clustering to detect planar structures in feature space based on the previously extracted straight line segments. Chen et al. [27] extracted planar patches by grouping points with similar features, measured by a local tangent plane distance, using an adaptive density-based spatial clustering of applications with noise (DBSCAN). Xie et al. [28] proposed to cluster points based on corrected normal vectors to fit a local quadratic surface (LQS) and then merge adjacent LQS to obtain the final plane segments. However, the results of feature clustering-based approaches highly rely on the quality of features and are also sensitive to the parameters of convergence criteria.

Energy optimization-based approaches [29], [30], [31], [32], [33] refine the initial plane segments generated by RANSAC, region growing, or feature clustering by optimizing the specifically designed energy functions. Yan et al. [29] applied the graph cuts [53] to minimize the global energy function based on multilabel optimization. However, the computational cost of multilabel optimization is expensive. To alleviate this issue, Li et al. [31] and Liu et al. [33] proposed to refine plane segments based on boundary relabeling. They first extracted coarse planar patches and then iteratively refined the boundary points. Yu and Lafarge [32] proposed to refine the coarse planar patches by minimizing a specifically designed multiobjective energy function with the use of five geometric operators. However, energy optimization-based approaches also need to design the convergence criteria without the prior information to terminate or guide the optimization of energy functions.

B. Machine Learning-Based Approaches

With resent advancements in machine learning for point cloud understanding, particularly in semantic classification [34], [35], [36] and instance segmentation [37], [38], [39], some machine learning-based approaches [40], [41], [42] have been proposed for roof plane segmentation. For example, Dey et al. [40] extracted seven geometric features (e.g., point curvature and normal deviation) and then fed them into two classic machine learning classifiers, namely support vector machine (SVM) and random forest (RF), to categorize roof points into boundary, fold, and planar points. This approach can facilitate the plane extraction and 3-D building reconstruction. Recently, some deep learning-based approaches have been proposed to leverage neural networks to automatically learn point features and achieve instance-aware plane segmentation. A multitask network [41] was proposed for roof plane segmentation. It projects each point into a high-dimensional feature space and ensures that the points included in the same plane instance have similar features. Then, the roof planes are segmented from the feature space. However, this method may fail to accurately segment boundaries. To solve this issue, Li et al. [42] proposed a boundary-aware clustering-based approach. They designed a three-branch network to first extract boundary points from point clouds and then cluster plane points in both Euclidean and embedding space to generate initial planar patches. Finally, they assigned the boundary points to their closest patches to refine the plane segments. Nevertheless, these approaches still cannot provide the critical prior information of the planes to guide the subsequent segmentation. They still rely on a point clustering algorithm with certain convergence criteria to segment roof planes.

C. Transformers

Transformer based on multihead attention mechanism, first proposed by Vaswani et al. [43], is capable of capturing long-distance dependencies from the input sequences. We have noticed that some transformer-based approaches for object detection [44], [45], [46], [47] and instance segmentation [48], [49], [50], [51] have been proposed and achieved good results in vision-related fields. For example, Carion et al. [44] proposed a transformer network, named DETR, for 2-D object detection. Based on a fixed number of learned queries for candidate objects, they enabled the DETR to directly output a set of bounding box predictions. After that, 3DETR [45] and Clusterformer [47] are proposed for indoor and outdoor 3-D object detection, respectively. Sun et al. [50] proposed an end-to-end network, named SPFormer, to perform 3-D instance segmentation. It first generates superpoints by grouping point features and then uses superpoint cross-attention to capture the instance information and further predict instance masks. OneFormer3 [51] is a unified transformer-based network designed for performing 3-D semantic, instance, and panoptic segmentation simultaneously. These transformer-based networks have shown that it is feasible and effective to use a query-based transformer for object detection or instance segmentation. Inspired by this, we consider to use a transformer architecture to detect the instance centers of all planes, thereby providing the prior information of the planes for subsequent roof plane segmentation and avoiding the need to design the complex and sensitive convergence conditions.

III. Proposed Method

In this article, we propose a novel transformer-based approach for roof plane segmentation from airborne LiDAR point clouds. As shown in Fig. 1, the proposed approach consists of two stages. In the first stage, we design a dual-branch transformer-based network to simultaneously predict the accurate center positions of planar patches and shift points closely toward their corresponding instance centers. In the second stage, we first utilize the shifted points around the predicted centers to estimate plane parameters and then assign the rest of the points to their nearest plane instances to generate the final planar patches.

A. Dual-Branch Transformer-Based Network

The pipeline of the proposed network is shown in Fig. 2. First, we generate query embeddings $\mathbf{Q}_{\text{emb}}$, namely initial centers, using the furthest point sampling method, and extract multiscale point-wise features $\mathbf{F}_{\text{enc}}$ using PointNet++ [52]

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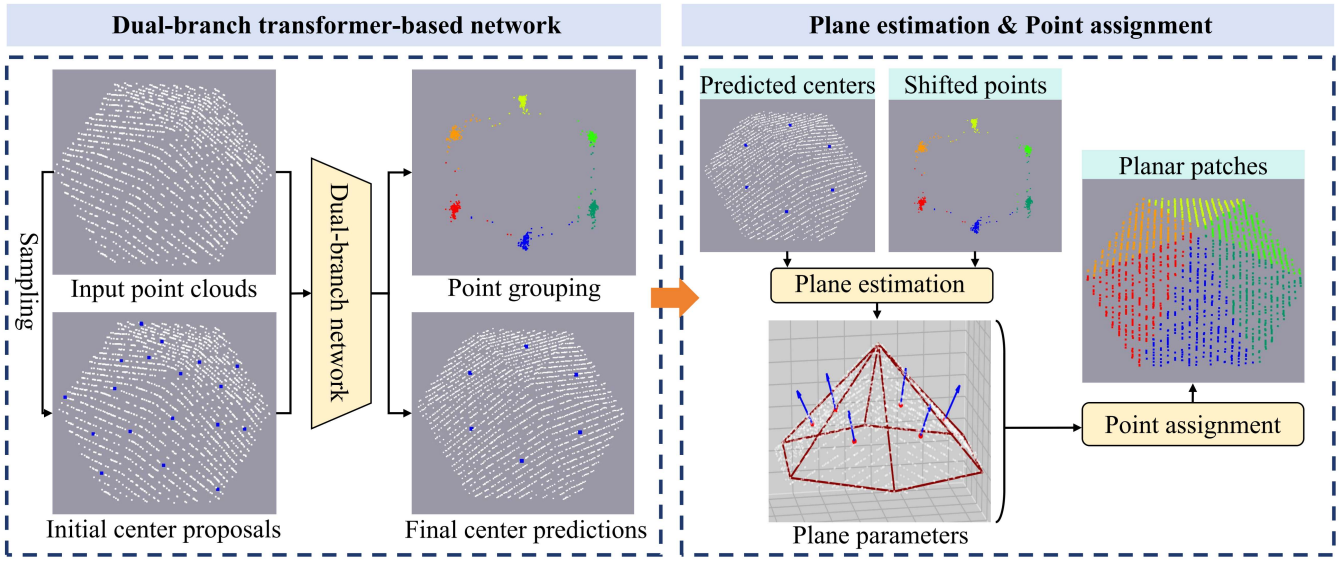


Fig. 1. Overall framework of the proposed roof plane segmentation approach.

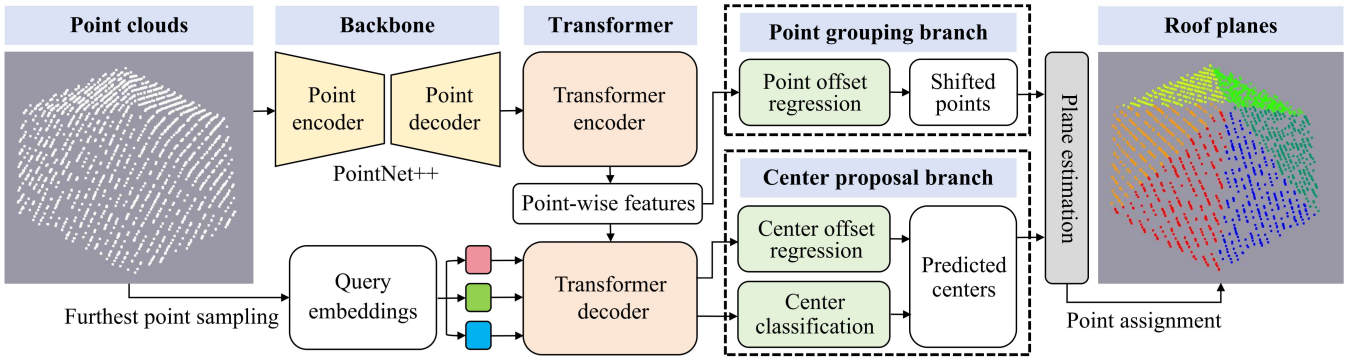


Fig. 2. Pipeline of the proposed dual-branch transformer-based network.

followed by a transformer encoder. Then, we design a dual-branch architecture to predict the plane instance centers and shift points toward their centers. We name these two branches as the center proposal branch and point grouping branch. In the center proposal branch, we input $\mathbf{Q}_{\text{emb}}$ and $\mathbf{F}_{\text{enc}}$ into the transformer decoder followed by two prediction heads to predict plane instance centers. In the point grouping branch, $\mathbf{F}_{\text{enc}}$ is used for point-wise offset regression to shift each point toward its instance center, thus deriving more points around the predicted centers that are robust for plane estimation.

1) Backbone Network: The 3-D point clouds with 3-D *XYZ* coordinates are taken as the input of our network. PointNet++ proposed by Qi et al. [52] is a classic network for point clouds feature extraction. Its advantages can be briefly summarized as follows: 1) it introduces a hierarchical local-to-global structure, consisting of set abstraction (SA) and feature propagation (FP) modules, to better capture fine-grained details in point clouds and 2) a multiscale grouping (MSG) strategy is applied for feature extraction and fusion, enabling strong adaptability to point clouds of varying densities. Given these strengths, we use PointNet++ as the backbone network to extract multiscale point-wise features $\mathbf{F}_{\text{bac}}$ with $d = 128$ dimensions.

2) Transformer Encoder: Considering that the attention mechanism of transformer can effectively capture long-range dependencies in input sequences, we apply the transformer encoder to further obtain more representative point-wise features through interpoint attention interactions. Our network uses standard transformer encoder blocks. As shown in Fig. 3(a), the transformer encoder layer includes a multihead self-attention module, a feedforward network (FFN), and normalization layers. For more details, we refer readers to [43]. Based on the features $\mathbf{F}_{\text{bac}}$ extracted by the backbone network, we utilize the transformer encoder to learn point-wise dependencies and generate features $\mathbf{F}_{\text{enc}}$ with $d = 128$ dimensions. Subsequently, $\mathbf{F}_{\text{enc}}$ is taken as inputs for the center proposal branch and the point grouping branch to predict planar centers and point-wise offsets, respectively.

3) Query Embeddings: We decide to use a set of learnable query embeddings to guide our transformer-based network to learn the information of plane instances. Specifically, we first sample K points as the initial center proposals, denoted as $\mathbf{C}_{\text{init}} = \{\mathbf{c}_i\}_{i=0}^{K}$, using the furthest points sampling method. Then, we apply the Fourier encoding method to generate K query embeddings $\mathbf{Q}_{\text{emb}}$ associated with the initial centers. Notably, compared to traditional positional encoding,

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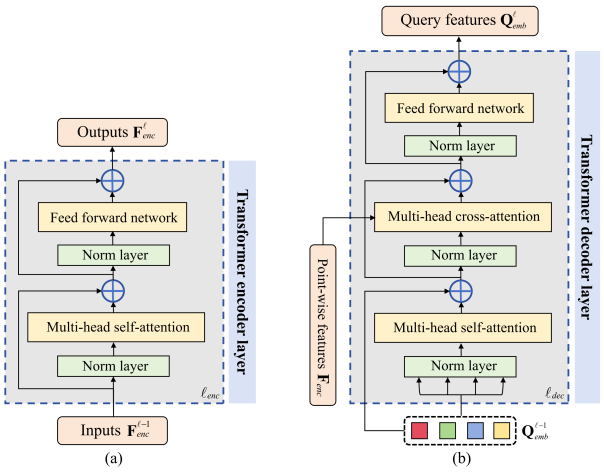


Fig. 3. Architecture of one layer of (a) transformer encoder and (b) transformer decoder. ℓ_{enc} and ℓ_{dec} denote the ℓth encoder and decoder layers, respectively. $\mathbf{F}^{\ell}_{\text{enc}}$ and $\mathbf{Q}^{\ell}_{\text{emb}}$ are the outputs of the ℓth encoder and decoder layers, respectively, and there are similar meanings for other symbols.

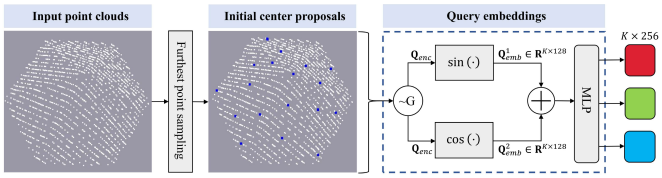


Fig. 4. Illustration of the query embeddings generation. "~G" refers to the multiply operation with a Gaussian matrix. "+" refers to the concatenation operation.

Fourier encoding is inherently more suitable for processing point clouds with sparse nature, as the orthogonality of its sine–cosine basis functions can effectively model missing regions in spatial domains [45]. As shown in Fig. 4, the *XYZ* coordinates of initial centers are multiplied by a Gaussian matrix to extract K encoded features $\mathbf{Q}_{\text{enc}}$ with $d = 128$ dimensions. Then, $\mathbf{Q}_{\text{enc}}$ is mapped to two Fourier positional embeddings $\mathbf{Q}^1_{\text{emb}} \in \mathbf{R}^{K\times 128}$ and $\mathbf{Q}^2_{\text{emb}} \in \mathbf{R}^{K\times 128}$ using sine and cosine transformations, respectively. Finally, $\mathbf{Q}^1_{\text{emb}}$ and $\mathbf{Q}^2_{\text{emb}}$ are concatenated and passed through an multi-layer perceptron (MLP) to generate the final query embeddings $\mathbf{Q}_{\text{emb}} \in \mathbf{R}^{K\times 256}$.

4) Transformer Decoder: To learn plane information from point clouds, the transformer decoder is employed to facilitate interactive information flow between query embeddings and point-wise features. Similar to the transformer encoder, we also use standard transformer decoder blocks [43] in our network. It should be noted that the transformer decoder layer employs two multihead attention modules, as shown in Fig. 3(b), which are applied for self-attention and cross-attention, respectively. In practice, we input K query embeddings $\mathbf{Q}^{\ell-1}_{\text{emb}}$ into the first multihead attention module of the ℓth decoder layer to obtain query-wise features $\mathbf{F}_{\text{que}}$ based on the self-attention mechanism. Then, we use the multihead cross-attention module to reason the interaction between $\mathbf{F}_{\text{que}}$ and $\mathbf{F}_{\text{enc}}$ extracted by the transformer encoder. Thus, we can integrate global context information into the queries. Finally, through an FFN, we generate K refined query embeddings $\mathbf{Q}^{\ell}_{\text{emb}}$ with $d = 256$ dimensions, which are further used for planar center predictions in the center proposal branch.

5) Center Proposal Branch: In this branch, we attempt to accurately predict the potential plane centers. To achieve that, we apply two prediction heads, which are essentially MLPs with different output dimensions, for center offset regression and center classification. Let $\mathbf{C}_{\text{truth}} = \{\mathbf{c}^{\text{gt}}_i\}^{M}_{i=0}$ denotes the ground truth centers, and M is the number of centers. We first utilize a center offset regression head to predict offset $\mathbf{o}^{\text{ct}}_i$ from each sampled initial center $\mathbf{c}_i$ to the associated plane center $\mathbf{c}^{\text{gt}}_i$. Then, by adding the corresponding $\mathbf{o}^{\text{ct}}_i$ to each $\mathbf{c}_i$, we can obtain the shifted centers, denoted as $\mathbf{C}_{\text{shift}} = \{\mathbf{c}^{\text{sh}}_i\}^{K}_{i=1}$, around the ground truth centers. Besides, to eliminate redundant center predictions, we use the confidence scores predicted by the center classification head to classify $\mathbf{C}_{\text{shift}}$ into two categories, namely positive or negative centers. Let $\mathbf{C}^{\text{pos}}_{\text{shift}} = \{\mathbf{c}^{\text{pos}}_j\}^{M}_{j=1}$ denote the positive centers, which are regarded as the final predicted planar centers. In Fig. 5(a)–(c), we visually illustrate the proposed center proposal branch.

6) Point Grouping Branch: Let $\mathbf{P} = \{\mathbf{p}_i\}^{N}_{i=0}$ denote the original point clouds, and N is the number of points. Because the space distribution of $\mathbf{P}$ is sparse, it may be difficult to directly find enough reliable original points near the predicted centers for the next plane estimation. To solve this problem, we decide to group coplanar points toward their plane instance centers. Specifically, we input the point-wise features $\mathbf{F}_{\text{enc}} \in \mathbf{R}^{N\times 128}$ into the point offset regression head to predict offset $\mathbf{o}^{\text{pt}}_i$ for each point $\mathbf{p}_i$. Then, we shift the points toward their associated centers using the predicted offsets. In this way, we can obtain the points that are closely distributed around the corresponding planar centers, as shown in Fig. 5(d) and (e).

B. Plane Estimation and Point Assignment

Based on the predicted planar centers and shifted points produced by the dual-branch transformer-based network, we perform plane estimation and point assignment to complete the final roof plane segmentation.

1) Plane Estimation: To fit a plane, at least three non-collinear points are required. As we know, the more robust points involved in the plane fitting, the more accurately the plane parameters can be estimated. However, due to the sparsity of the point clouds, there may not be enough robust points around the predicted centers. This is the motivation behind our design of the point grouping branch, i.e., to shift coplanar points toward the corresponding plane center, thereby facilitating the plane estimation. Specifically, we first find the k nearest shifted points around the predicted positive center $\mathbf{c}^{\text{pos}}_j$, as shown in Fig. 6(b). Then, we apply the least-squares fitting method [55] to estimate the planar parameters using the original points associated with the k shifted points. Let $\mathcal{P}_{\text{est}} = \{\mathcal{P}_j\}^{M}_{j=1}$ denotes the M estimated planes, while $\mathbf{c}^{\text{pos}}_j$ represents the planar center of $\mathcal{P}_j$. The visualization of the estimated planes is shown in Fig. 6(c).

2) Point Assignment: We have already obtained the predicted planar centers and estimated planar parameters, which are the critical prior information for plane segmentation. Thus, we can directly assign the points to their nearest planes to extract plane segments, avoid designing complex and sensitive

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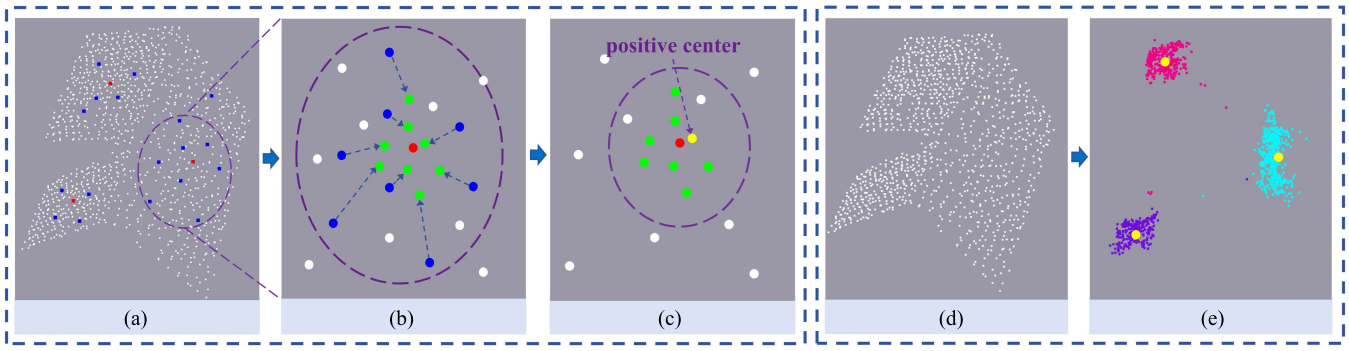


Fig. 5. Illustration of the dual-branch network. (a) Initial centers. (b) Center offset regression. (c) Center classification. (d) Input points. (e) Point groups. In (a)–(d), the white, blue, green, red, and yellow dots represent the input points, the initial sampled centers, the shifted centers, the ground truth plane center, and the final positive center, respectively. The blue arrows represent the offsets. In (e), the yellow dots represent the predicted positive centers and the dots with the other three colors represent the three point groups generated by the point grouping branch.

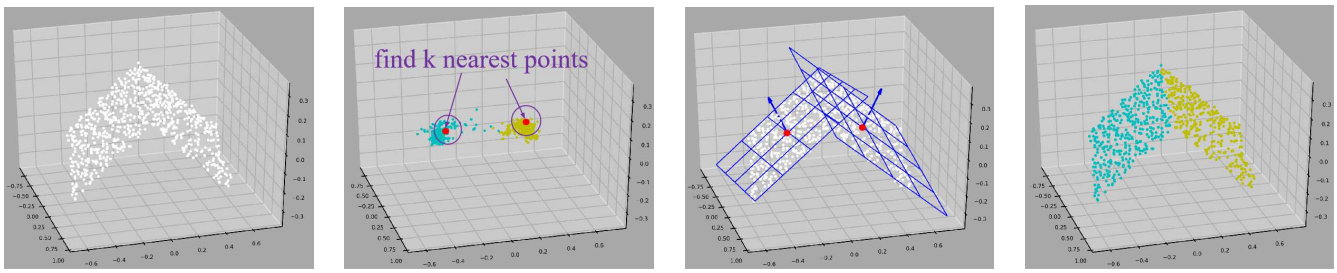


Fig. 6. Illustration of the plane estimation and point assignment. The white dots in (a) and (c) represent the initial point clouds. In (b), the cyan and yellow dots represent the shifted points of two planar patches, and the red dots represent the two predicted planar centers. In (c), the blue wireframes and arrows represent the estimated planes and normal vectors. In (d), the cyan and yellow dots represent the final two roof plane segments.

convergence criteria. To achieve that, we propose a simple yet effective point assignment method to assign the original unlabeled points, represented by the white dots in Fig. 6(c), to their nearest estimated planes. Specifically, we design a hybrid distance metric that considers both point-to-center and point-to-plane distance to guide the assignment of each point. The formulation of the hybrid distance metric is defined as follows:

$$\text{Dist}\left(\mathbf{p}_i, \mathcal{P}_j\right) = \frac{D_{\text{ct}}\left(\mathbf{p}_i, \mathbf{c}_j^{\text{pos}}\right)}{\sum_{j=1}^{M} D_{\text{ct}}\left(\mathbf{p}_i, \mathbf{c}_j^{\text{pos}}\right)} + \frac{D_{\text{pl}}\left(\mathbf{p}_i, \mathcal{P}_j\right)}{\sum_{j=1}^{M} D_{\text{pl}}\left(\mathbf{p}_i, \mathcal{P}_j\right)} \quad (1)$$

where $\text{Dist}(\mathbf{p}_i, \mathcal{P}_j)$ refers to the hybrid distance between the ith unlabeled point $\mathbf{p}_i$ and the jth estimated plane $\mathcal{P}_j$. $\mathbf{c}_j^{\text{pos}}$ denotes the predicted positive center of $\mathcal{P}_j$, while $D_{\text{ct}}(\cdot)$ and $D_{\text{pl}}(\cdot)$ represent the point-to-center and point-to-plane $L1$ norm distances, respectively. To balance the scale differences between the two distances, we normalize them individually before summing them up. In practice, we assign each point $\mathbf{p}_i$ to the estimated plane with the minimum hybrid distance, thereby obtaining the final plane segments, as shown in Fig. 6(d). Notably, compared to relying only on a single distance constraint, this hybrid distance-based point assignment strategy can effectively distinguish nearly coplanar patches and accurately segment boundary points, as detailed in Section IV-D, thus achieving the optimal segmentation results.

C. Bipartite Matching and Loss Function

1) Bipartite Matching: To reduce the redundant planar center predictions, inspired by [44], we apply a bipartite matching method to assign the optimal predicted center for each ground truth center. Specifically, we first match K shifted centers to M ground truth centers, resulting in $K \times M$ matching center pairs. Then, for each matching pair $(\mathbf{c}_i^{\text{sh}}, \mathbf{c}_i^{\text{gt}})$, we design a cost function $f_{\text{cost}}(\mathbf{c}_i^{\text{sh}}, \mathbf{c}_i^{\text{gt}})$ to simultaneously measure their distance and semantic matching cost, which can be formulated as

$$f_{\text{cost}}\left(\mathbf{c}_i^{\text{sh}}, \mathbf{c}_i^{\text{gt}}\right) = \left\|\mathbf{c}_i^{\text{sh}} - \mathbf{c}_i^{\text{gt}}\right\|_1 - \mathbf{s}_{\mathbf{c}_i^{\text{sh}}}^{\text{pos}} \quad (2)$$

where $\|\cdot\|_1$ refers to the $L1$ norm distance between the shifted center $\mathbf{c}_i^{\text{sh}}$ and the ground truth center $\mathbf{c}_i^{\text{gt}}$ and $\mathbf{s}_{\mathbf{c}_i^{\text{sh}}}^{\text{pos}}$ refers to the predicted confidence score of the positive label associated with $\mathbf{c}_i^{\text{sh}}$.

Then, we apply the Hungarian algorithm [54] to find the final M optimal matching pairs between all the shifted centers and ground truth centers. Based on this, the center loss is calculated to supervise the model in predicting the most reliable planar centers.

2) Loss Function: To train our dual-branch transformer-based network, we design a loss function $\mathcal{L}_{\text{total}}$, which consists of three components: the point offset regression loss $\mathcal{L}_{\text{pt_reg}}$, the center offset regression loss $\mathcal{L}_{\text{ct_reg}}$, and the center classification loss $\mathcal{L}_{\text{ct_cls}}$. We formulate $\mathcal{L}_{\text{total}}$ as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pt_reg}} + \mathcal{L}_{\text{ct_reg}} + \mathcal{L}_{\text{ct_cls}}. \quad (3)$$

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$\mathcal{L}_{\text{pt_reg}}$ constrains that the predicted point-wise offsets are the same as the ground truth offsets. Combining both distance and direction constraints, similar to [37], we calculate $\mathcal{L}_{\text{pt_reg}}$ as

$$\mathcal{L}_{\text{pt_reg}} = \sum_{i=1}^{N} \left(\left\| \mathbf{o}_i^{\text{pt}} - \left(\mathbf{c}_{\mathbf{p}_i}^{\text{gt}} - \mathbf{p}_i \right) \right\|_1 + \frac{\mathbf{o}_i^{\text{pt}}}{\left| \mathbf{o}_i^{\text{pt}} \right|} \cdot \frac{\mathbf{c}_{\mathbf{p}_i}^{\text{gt}} - \mathbf{p}_i}{\left| \mathbf{c}_{\mathbf{p}_i}^{\text{gt}} - \mathbf{p}_i \right|} \right) \quad (4)$$

where $\mathbf{o}_i^{\text{pt}}$ refers to the predicted offset of $\mathbf{p}_i$ and $\mathbf{c}_{\mathbf{p}_i}^{\text{gt}}$ refers to the ground truth center associated with $\mathbf{p}_i$. The subtraction of $\mathbf{c}_{\mathbf{p}_i}^{\text{gt}}$ and $\mathbf{p}_i$ refers to the ground truth offset. Notably, the first term calculates the $L1$ norm distance between the predicted and ground truth offset, while the second term calculates the cosine similarity between them to measure their directional deviations.

We compute $\mathcal{L}_{\text{ct_reg}}$ and $\mathcal{L}_{\text{ct_cls}}$ based on the matched pairs between shifted centers and ground truth centers. $\mathcal{L}_{\text{ct_reg}}$ constrains offset regression of initial centers, ensuring that the predicted offsets are the same as the ground truth center offset. The formulation of $\mathcal{L}_{\text{ct_reg}}$ is defined similar to $\mathcal{L}_{\text{pt_reg}}$ as follows:

$$\mathcal{L}_{\text{ct_reg}} = \sum_{i=1}^{M} \left(\left\| \mathbf{o}_i^{\text{ct}} - \left(\mathbf{c}_{\mathbf{c}_i}^{\text{gt}} - \mathbf{c}_i \right) \right\|_1 + \frac{\mathbf{o}_i^{\text{ct}}}{\left| \mathbf{o}_i^{\text{ct}} \right|} \cdot \frac{\mathbf{c}_{\mathbf{c}_i}^{\text{gt}} - \mathbf{c}_i}{\left| \mathbf{c}_{\mathbf{c}_i}^{\text{gt}} - \mathbf{c}_i \right|} \right) \quad (5)$$

where $\mathbf{o}_i^{\text{ct}}$ refers to the predicted center offset of $\mathbf{c}_i$ and $\mathbf{c}_{\mathbf{c}_i}^{\text{gt}}$ refers to the ground truth center associated with $\mathbf{c}_i$. The subtraction of $\mathbf{c}_{\mathbf{c}_i}^{\text{gt}}$ and $\mathbf{c}_i$ refers to the ground truth center offset. The first and second terms are computed as distance and direction constraints between the predicted center offset and ground truth, respectively.

$\mathcal{L}_{\text{ct_cls}}$ is designed for training the network to suppress the redundant center predictions. To achieve that, we ensure that the shifted centers that do not match with the ground truth are classified as negative centers, while the matched shifted centers are classified as positive centers. We apply the standard binary cross-entropy loss $\mathcal{L}_{\text{bce}}(\cdot)$ [56] for $\mathcal{L}_{\text{ct_cls}}$, which can be defined as

$$\mathcal{L}_{\text{ct_cls}} = \sum_{i=1}^{K} \mathcal{L}_{\text{bce}} \left(\mathbf{s}_{\mathbf{c}_i^{\text{sh}}}, l_{\text{cls}} \left(\mathbf{c}_i^{\text{sh}} \right) \right) \quad (6)$$

where $\mathbf{s}_{\mathbf{c}_i^{\text{sh}}}$ and $l_{\text{cls}}(\mathbf{c}_i^{\text{sh}})$ refer to the predicted confidence score and the label of each shifted center $\mathbf{c}_i^{\text{sh}}$, respectively. It should be noted that only the M matched shifted centers are involved in the $\mathcal{L}_{\text{ct_reg}}$ calculation, while $\mathcal{L}_{\text{ct_cls}}$ computes the total classification loss for all K shifted centers. In addition, during the inference phase, we use the predicted value of confidence score to determine the category of each predicted center. Specifically, centers with scores higher than 0.5 are considered as the final predicted positive centers.

IV. Experimental Results

A. Implementation Details

1) Network Architecture Details: We choose PointNet++ [52] as the backbone to extract initial point-wise features. We utilize four multiscale SA (MSA) modules, with receptive radii of $[0.05, 0.1]$, $[0.1, 0.2]$, $[0.2, 0.4]$, and $[0.4, 0.8]$, to downsample and extract multiscale features. Then, four FP modules are applied to upsample and interpolate to acquire the point-wise features with 128 dimensions.

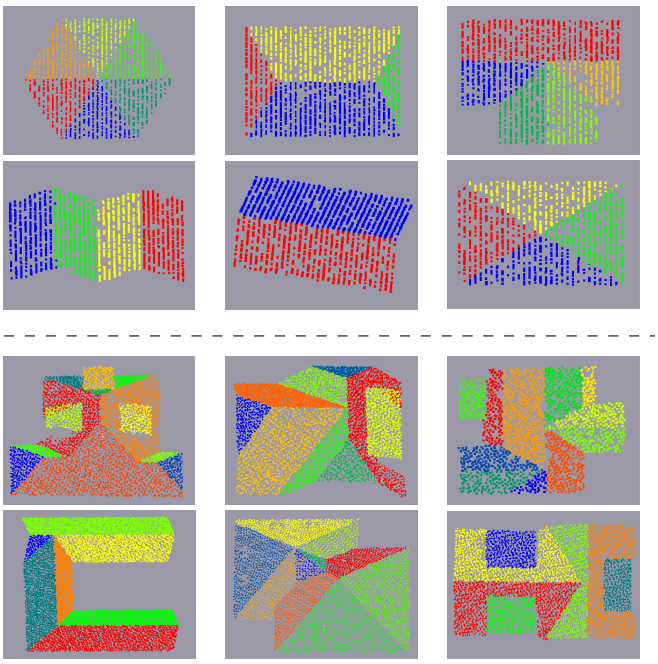


Fig. 7. Visualization of several roof samples in the Roofpc3D and Building3D datasets. For each roof sample, the dots in different colors represent different planar patches.

The transformer encoder has two layers, and each layer has a self-attention module with four heads followed by an FFN. The transformer decoder has eight layers, and each layer has two attention modules with four heads and an FFN. Similar to [45], we use the standard normalization layers. The dropout of encoder and decoder layers is set as 0.1 and 0.3, respectively.

In the center proposal branch, center offset regression and classification heads are two MLPs with $d = 3$ and $d = 2$ output dimensions. The input and hidden dimensions are 256 and [128, 64], respectively. Similarly, an MLP with $d = 3$ output dimensions is applied for the point offset regression head in the point grouping branch.

2) Training Details: The network is trained on a single NVIDIA RTX 3090 GPU under the PyTorch framework. We apply AdamW [57] as optimizer, with a weight decay of 0.1. The learning rate is declined to 1×10^{-6} by a cosine schedule. We train the network for 180 epochs, with the batch size of 30. The query numbers for two datasets are set to 16 and 32.

B. Dataset Preparation

1) Roofpc3D Dataset: This is a synthetic dataset for roof plane segmentation [42]. It consists of 14 roof types, with the number of planes in each type varying between 2 and 8. In addition, each roof type is composed of 1000 training samples and 100 testing samples. Based on this, we divided this dataset into training and testing sets with a ratio of 10:1, resulting in 14000 samples for training and 1400 for testing. There are several roof building samples, as shown in Fig. 7 (top two lines).

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TABLE I
QUANTITATIVE EVALUATION OF ALL APPROACHES ON TWO DATASETS

Approaches	Roofpc3D dataset				Building3D dataset			
	mCov	mWCov	mPrec	mRec	mCov	mWCov	mPrec	mRec
Region growing	0.6777	0.7274	0.6991	0.8406	0.7383	0.7820	0.7061	0.8163
RANSAC	0.7079	0.7402	0.6092	0.8198	0.7768	0.8269	0.8287	0.8389
PointGroup	0.7998	0.8379	0.8789	0.9067	0.7716	0.8507	0.8470	0.8257
GoCoPP	0.8377	0.8479	0.7749	0.9271	0.8012	0.8437	0.8400	0.8695
Liu et al.'s approach	0.8525	0.8819	0.9609	0.9233	0.8157	0.8675	0.9262	0.8571
Our approach	**0.9376**	**0.9544**	**0.9952**	**0.9850**	**0.9007**	**0.9402**	**0.9803**	**0.9392**

2) Building3D Dataset: In our previous work [42], we constructed a real urban-scale building dataset, which contains 18227 roof samples with ground truth plane labels. In this article, we further expanded this dataset to 18708 samples. Besides, we manually corrected erroneous plane labels in some roof samples to further refine this dataset. We also divided this dataset into training and testing sets with a ratio of 10:1. Notably, because of the more complex roof types and the increased number of planes, this dataset is challenging for roof plane segmentation. We present several roof samples of the Building3D dataset in Fig. 7 (bottom two lines).

C. Evaluation Metrics

To quantitatively evaluate the experimental results of plane segmentation, we adopt four instance-level evaluation metrics: mean coverage (mCov), mean weighted coverage (mWCov), mean precision (mPrec), and mean recall (mRec). For more details, please refer to our previous work [42].

D. Comparative Experiments

To validate the effectiveness of the proposed plane segmentation approach, we conducted several comparative experiments. Specifically, we compared the segmentation results of four traditional approaches, including region growing [2], RANSAC [11], GoCoPP [32], and Liu et al.'s approach [33], as well as a deep learning-based approach, PointGroup [37], with our approach on Roofpc3D and Building3D testing datasets. We apply the Open3D[1] and PCL[2] libraries for the implementation of region growing and RANSAC, respectively. The source codes of GoCoPP, PointGroup, and Liu et al.'s approach [33] are provided by their authors. The quantitative evaluation results of all approaches on two testing datasets are reported in Table I.

From Table I, it can be concluded that our approach significantly outperforms the other five approaches for all four evaluation metrics on two datasets. On the Roofpc3D dataset, except for our approach, the best results of mCov and mWCov are achieved by Liu et al.'s approach [33], with scores of 0.8525 and 0.8819, respectively. In comparison, our approach achieves scores of 0.9376 on mCov and 0.9544 on mWCov, each showing an improvement of at least seven percentage points. The high scores on these two metrics indicate that our approach is accurate in label assignment for each planar point. Additionally, for the last two metrics, mPrec and mRec, our approach also achieves the highest scores of 0.9952 and 0.9850, respectively, which demonstrates that our approach performs well in instance-level plane segmentation. On the more challenging Building3D dataset, despite a slight decline in the results of all four metrics, our approach still outperforms all other approaches, achieving the best performance.

To qualitatively compare and evaluate the roof plane segmentation results of different approaches, we select some roof samples from two testing datasets and visualize their segmentation results, as shown in Figs. 8–10. Given the synthetic dataset, in Fig. 8(b1) and (b2), we present the results of region growing. It fails to segment the boundary points between the adjacent patches and generates oversegmented patches. RANSAC produces some spurious patches, as shown in Fig. 8(c1), and causes undersegmentation, as shown in Fig. 8(c2). The results of PointGroup show incorrect boundary segments and undersegmented patches, as shown in Fig. 8(d1) and (d2). We observe that GoCoPP and Liu et al.'s approach also cannot provide correct and smooth boundary segments. Besides, GoCoPP may overly segment planar patches, as shown in Fig. 8(e1), while Liu et al.'s approach produces the undersegmented patches, as shown by the blue wireframes in Fig. 8(f2). In comparison, our approach achieves the best segmentation results that are largely consistent with the ground truth for each planar patch.

In Fig. 9, it can be found that the best results are also achieved by our approach on the Building3D testing dataset. Region growing still fails in boundary points segmentation and struggles with the issue of oversegmentation. RANSAC continues to generate spurious patches, as shown in Fig. 9(c1), and fails to distinguish between two approximately coplanar patches, as shown in Fig. 9(c2). PointGroup also fails to assign the correct labels to the boundary points and produces undersegmented patches. GoCoPP and Liu et al.'s approach still cannot segment boundary points smoothly and also cannot avoid oversegmentation. Besides, Liu et al.'s approach incorrectly merges two coplanar patches into one patch, as shown in Fig. 9(f2). To further validate the effectiveness of our approach, we also select some roof samples with more complex structures, each comprising over ten planar patches, as illustrated in Fig. 10. Evidently, as the number of planes increases, other comparative approaches exhibit more

[1]Available at https://www.open3d.org/
[2]Available at https://github.com/PointCloudLibrary/pcl/

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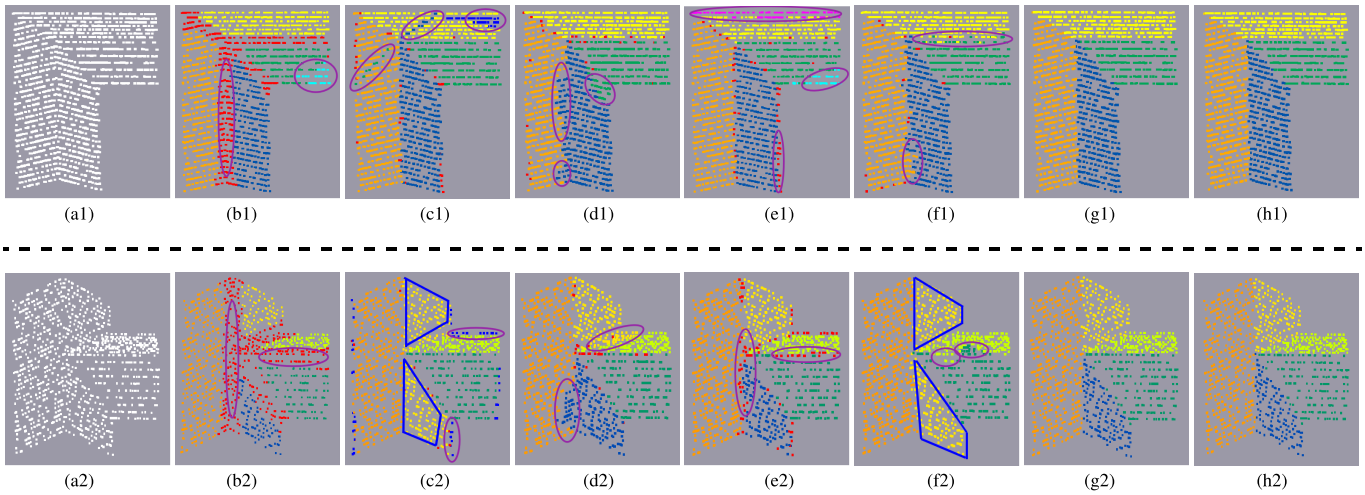


Fig. 8. Visualization of the roof segmentation results of all approaches on the Roofpc3D dataset. The purple ellipses highlight some areas with segmentation errors. The red dots represent the unlabeled points or outliers. The blue wireframes highlight the undersegmented patches. The dots in other colors represent different patches. (a1) and (a2) Original input. (b1) and (b2) Region growing. (c1) and (c2) RANSAC. (d1) and (d2) PointGroup. (e1) and (e2) GoCoPP. (f1) and (f2) Liu et al.'s. (g1) and (g2) Our approach. (h1) and (h2) Ground truth.

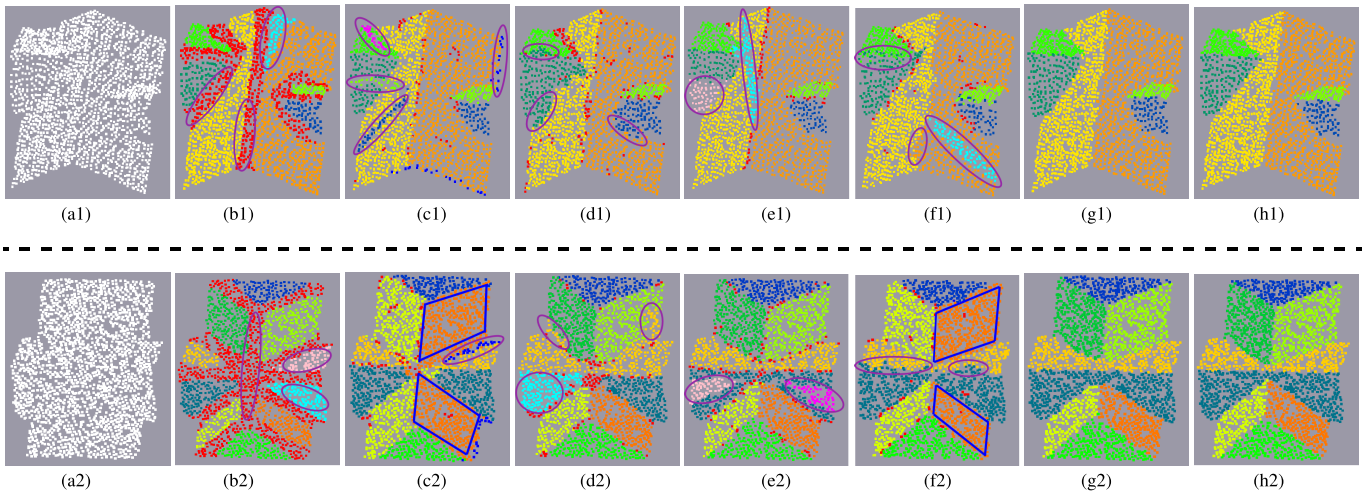


Fig. 9. Visualization of the roof segmentation results of all approaches on the Building3D dataset. The purple ellipses highlight some areas with segmentation errors. The red dots represent the unlabeled points or outliers. The blue wireframes highlight the undersegmented patches. The dots in other colors represent different patches. (a1) and (a2) Original input. (b1) and (b2) Region growing. (c1) and (c2) RANSAC. (d1) and (d2) PointGroup. (e1) and (e2) GoCoPP. (f1) and (f2) Liu et al.'s. (g1) and (g2) Our approach. (h1) and (h2) Ground truth.

severe undersegmentation or oversegmentation issues, especially failing to segment smaller planar patches. In contrast, our approach can accurately detect the number of planes in the point clouds and perform high-quality segmentation, as shown in the Fig. 10(g1) and (g2). Furthermore, given that the Building3D dataset is an urban-scale real-world benchmark, we present the large-scale roof segmentation results in Fig. 11, which covers all 1824 test samples in the test set.

E. Ablations

Our approach contains two main stages. In the first stage, we directly predict potential planar centers from point clouds with the proposed network. In the second stage, we perform plane segmentation based on plane estimation and point assignment. To further validate the effectiveness of the two stages and understand their influence on our approach, we conducted several ablation experiments on two datasets.

1) Effects of Varied Numbers of Transformer Decoder Layers and Initial Center Queries: Notably, each layer of the transformer decoder in our network is trained to output center predictions followed by the shared center proposal branch, and all the predictions are involved in the center loss calculation. Thus, the change in the number of decoder layers can affect the accuracy of center predictions, which in turn impacts the final plane segmentation. Besides, the number of the initial center queries is also a key factor influencing the center predictions. In our experiments, we set the number of decoder layers to 2, 4, 6, and 8, with the number of queries set to 16, 32, and 64 for each case. The quantitative evaluation results are shown in Table II. For clarity, we have visualized the results of mCov on two datasets in Fig. 12.

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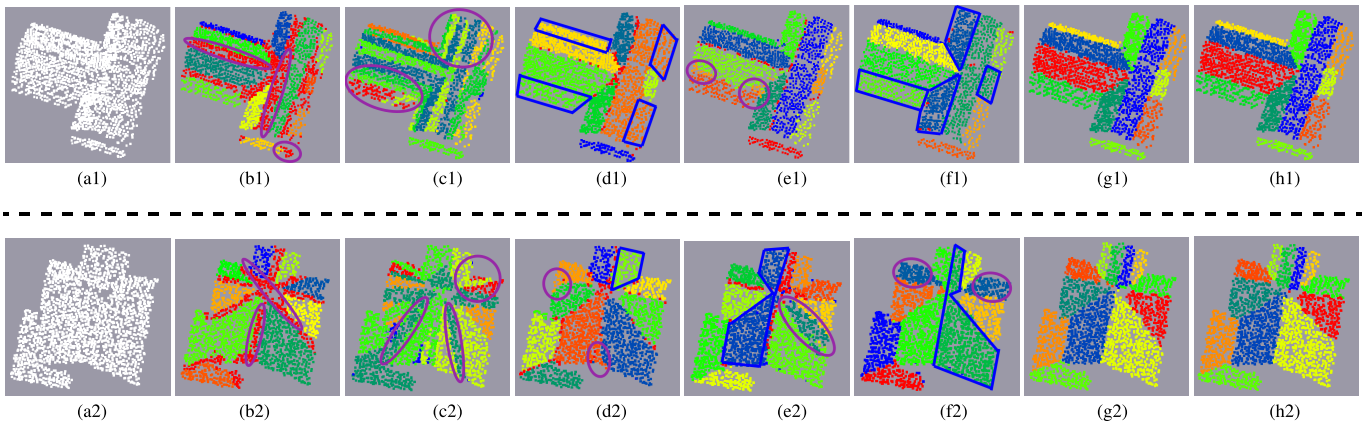


Fig. 10. Visualization of the roof segmentation results of all approaches on the Building3D dataset (more challenging roof samples). The purple ellipses highlight some areas with segmentation errors. The red dots represent the unlabeled points or outliers. The blue wireframes highlight the undersegmented patches. The dots in other colors represent different patches. (a1) and (a2) Original input. (b1) and (b2) Region growing. (c1) and (c2) RANSAC. (d1) and (d2) PointGroup. (e1) and (e2) GoCoPP. (f1) and (f2) Liu et al.'s. (g1) and (g2) Our approach. (h1) and (h2) Ground truth.

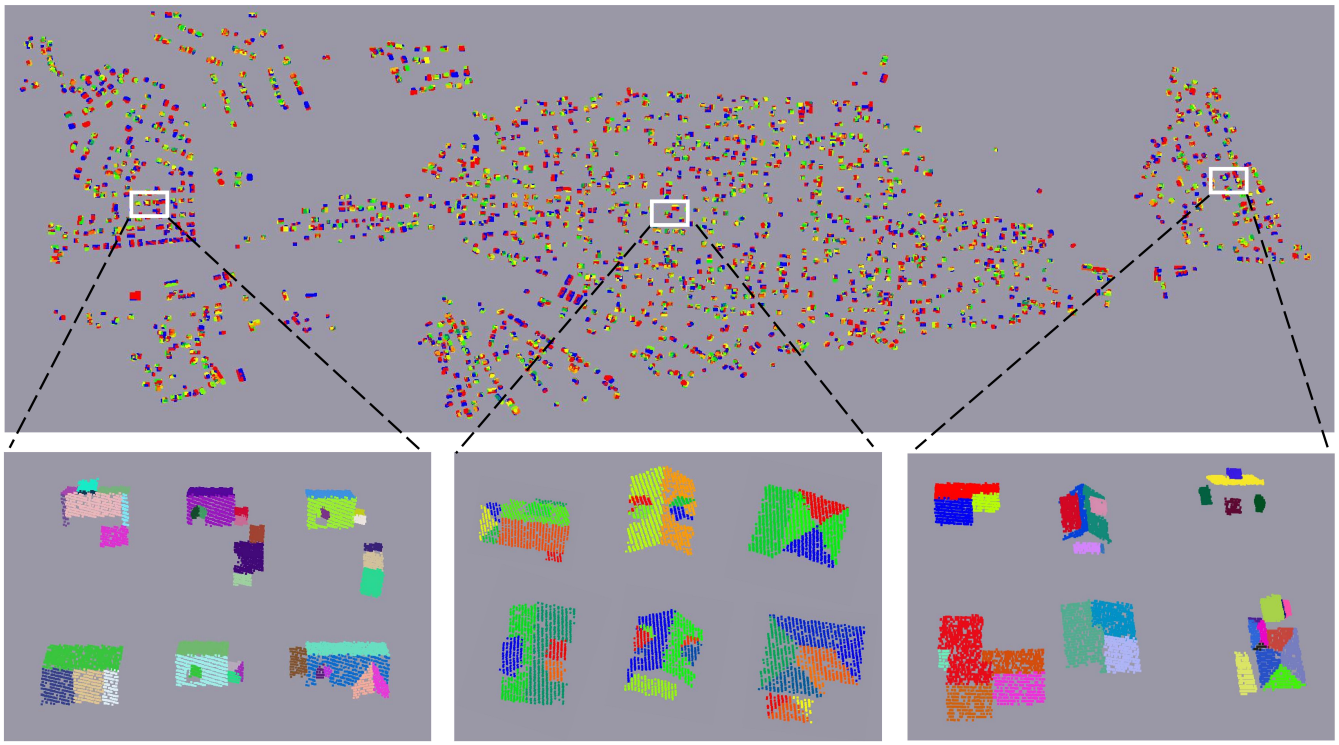


Fig. 11. Illustration of the large-scale roof segmentation results of the Building3D dataset. The first row displays an overview of the segmentation results for all 1824 building roofs in the test set, while the second row highlights three local regions.

As the number of decoder layers increases, the value of mCov gradually grows for both datasets. The overall improvement from $l = 2$ to $l = 8$ is approximately 1% on the Roofpc3D dataset and 4% on the Building3D dataset. However, the growth rate of mCov decreases significantly on both datasets. Considering that further increasing the number of decoder layers leads to a substantial growth in computational cost, as shown in Table III, we set the default number of decoder layers to 8. Besides, compared to the number of decoder layers, we observed that the number of initial queries has a smaller impact on mCov. The best results of mCov are achieved with a query number of 16 on the Roofpc3D dataset and 32 on the Building3D dataset. This is because the Building3D dataset has more complex roof types and a larger number of roof planes. In addition, similar conclusions can be drawn for the other three evaluation metrics from Table II.

2) Effects of Varied Number of Shifted Points for Plane Estimation and Different Distance Metrics for Point Assignment: In the second stage of our approach, we first find k nearest shifted points around each predicted center for plane estimation. Then, we design a hybrid distance metric, considering both point-to-plane and point-to-center distance, to guide point assignment. To evaluate the effect of the varied value of k and different distance metrics on our approach,

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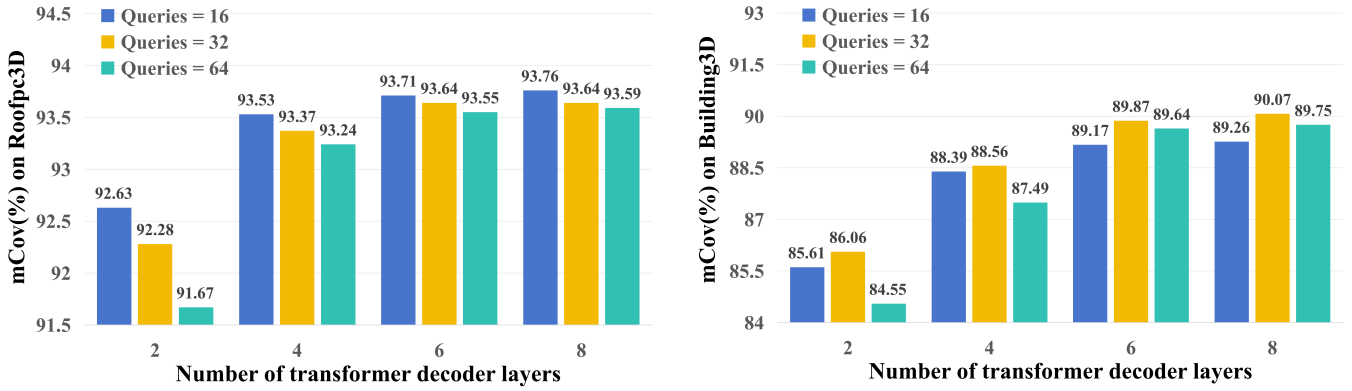


Fig. 12. Visualization of the results of mCov on two datasets derived from Table II.

TABLE II

QUANTITATIVE EVALUATION OF OUR APPROACH WITH VARIED NUMBERS OF TRANSFORMER DECODER LAYERS AND INITIAL CENTER QUERIES ON TWO DATASETS

Number of queries		**Queries = 16**				**Queries = 32**				**Queries = 64**			
Decoder layers		$l=2$	$l=4$	$l=6$	$l=8$	$l=2$	$l=4$	$l=6$	$l=8$	$l=2$	$l=4$	$l=6$	$l=8$
Roofpc3D dataset	mCov	0.9263	0.9353	0.9371	**0.9376**	0.9229	0.9373	0.9364	0.9364	0.9168	0.9324	0.9355	0.9359
	mWCov	0.9441	0.9543	0.9537	**0.9544**	0.9398	0.9543	0.9540	0.9529	0.9350	0.9510	0.9521	0.9494
	mPrec	0.9898	0.9962	0.9947	**0.9952**	0.9908	0.9938	0.9925	0.9949	0.9872	0.9951	0.9972	0.9941
	mRec	0.9716	0.9840	0.9848	**0.9850**	0.9701	0.9828	0.9838	0.9844	0.9620	0.9841	0.9859	0.9839
Building3D dataset	mCov	0.8561	0.8839	0.8917	0.8926	0.8606	0.8856	0.8987	**0.9007**	0.8455	0.8749	0.8964	0.8975
	mWCov	0.9013	0.9252	0.9324	0.9323	0.9052	0.9261	0.9387	**0.9402**	0.8939	0.9191	0.9357	0.9369
	mPrec	0.9575	0.9679	0.9742	0.9718	0.9609	0.9717	0.9801	**0.9803**	0.9570	0.9711	0.9781	0.9786
	mRec	0.8957	0.9297	0.9342	0.9353	0.9020	0.9295	0.9380	**0.9392**	0.8864	0.9171	0.9366	0.9391

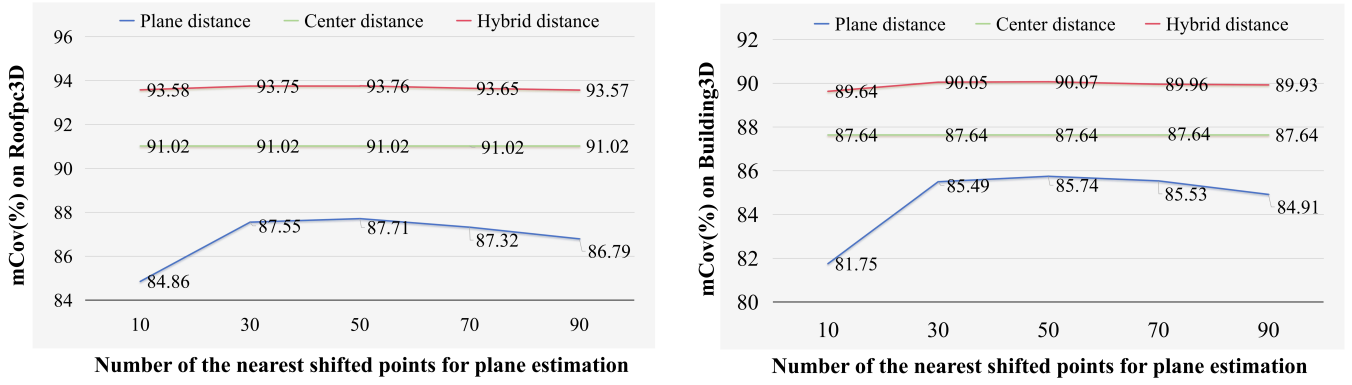


Fig. 13. Visualization of the results of mCov on two datasets derived from Table IV.

TABLE III

COMPUTATIONAL COST WITH VARIED NUMBERS OF TRANSFORMER DECODER LAYERS. #PARA. REFERS TO THE PARAMETERS OF THE NETWORK. FLOPS REFERS TO THE NUMBER OF FLOATING-POINT OPERATIONS PER SECOND IN A FORWARD PASS

Decoder layers	$l=2$	$l=4$	$l=6$	$l=8$
# Para. (MB)	14.2195	19.2508	24.2820	29.3133
FLOPs (GB)	18.8075	21.6360	24.4645	27.2930

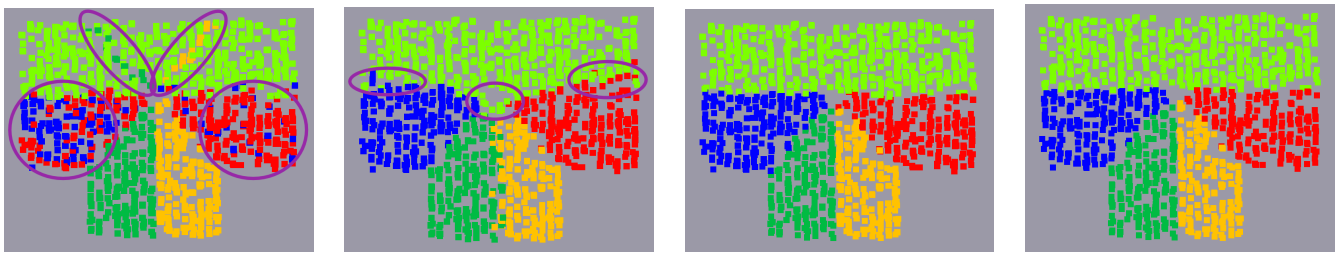
we conduct relevant ablation experiments on two datasets. Specifically, we set k to 10, 30, 50, 70, and 90 for plane estimation, each followed by applying three distance metrics for point assignment. The evaluation results are shown in Table IV. For a clear illustration, we have visualized the results of mCov on two datasets in Fig. 13.

We found that using point-to-plane distance for point assignment is sensitive to the value of k. This is because the value of k can directly affect the quality of the estimated planes. When k is small, the distribution of the shifted points may become overly concentrated. When k is large, shifted points may include points from other planes, namely outliers. These two situations both lead to a lack of robustness in plane estimation. In addition, this distance metric achieves the lowest score of mCov on two datasets because it cannot effectively distinguish

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TABLE IV
QUANTITATIVE EVALUATION OF OUR APPROACH WITH VARIED NUMBERS OF SHIFTED POINTS FOR PLANE ESTIMATION AND THREE DISTANCE METRICS FOR POINT ASSIGNMENT ON TWO DATASETS

Datasets	k-nearest shifted points	Plane distance				Center distance				Hybrid distance			
		mCov	mWCov	mPrec	mRec	mCov	mWCov	mPrec	mRec	mCov	mWCov	mPrec	mRec
Roofpc3D	$k = 10$	0.8486	0.8732	0.9386	0.9347	0.9102	0.9353	0.9903	0.9747	0.9358	0.9529	0.9940	0.9834
	$k = 30$	0.8755	0.8970	0.9568	0.9535	0.9102	0.9353	0.9903	0.9747	0.9375	0.9543	0.9941	0.9839
	$k = 50$	0.8771	0.8999	0.9488	0.9464	0.9102	0.9353	0.9903	0.9747	**0.9376**	**0.9544**	**0.9952**	**0.9850**
	$k = 70$	0.8730	0.9005	0.9405	0.9366	0.9102	0.9353	0.9903	0.9747	0.9365	0.9542	0.9935	0.9832
	$k = 90$	0.8679	0.8994	0.9260	0.9217	0.9102	0.9353	0.9903	0.9747	0.9357	0.9541	0.9934	0.9827
Building3D	$k = 10$	0.8175	0.8648	0.9314	0.8999	0.8764	0.9256	0.9743	0.9250	0.8964	0.9360	0.9786	0.9374
	$k = 30$	0.8549	0.8998	0.9581	0.9253	0.8764	0.9256	0.9743	0.9250	0.9005	0.9399	0.9800	0.9391
	$k = 50$	0.8574	0.9027	0.9581	0.9244	0.8764	0.9256	0.9743	0.9250	**0.9007**	**0.9402**	**0.9803**	**0.9392**
	$k = 70$	0.8553	0.9029	0.9543	0.9200	0.8764	0.9256	0.9743	0.9250	0.8996	0.9400	0.9801	0.9382
	$k = 90$	0.8491	0.9008	0.9464	0.9105	0.8764	0.9256	0.9743	0.9250	0.8993	0.9400	0.9799	0.9379



(a) (b) (c) (d)

Fig. 14. Segmentation results of one roof example with different distance metrics for point assignment (k = 50). The purple ellipses highlight some areas with segmentation errors. The dots with varied colors represent the different planar patches. (a) w/ plane distance. (b) w/ center distance. (c) w/ hybrid distance. (d) ground truth.

approximately coplanar or implicit intersecting planar patches, as shown in Fig. 14(a). By contrast, using point-to-center distance can offer better results and be completely unaffected by the value of k. However, it may still fail to segment boundary points correctly, as shown in Fig. 14(b). This is because adjacent patches generally have different shapes and sizes. The boundary points of a larger patch may be incorrectly assigned to the center of a smaller patch by only following the point-to-center distance. The hybrid distance metric, considering both distance metrics, mitigates the aforementioned issues and achieves the highest score of mCov on two datasets, as shown in Fig. 13. Besides, it is also not sensitive to the value of k, thus reducing the dependency of our approach on hyperparameters. Similar results can be found with the other three evaluation metrics in Table IV. In our experiments, we empirically set k = 50 for two datasets.

V. CONCLUSION

In this article, we develop a novel transformer-based approach for roof plane segmentation from airborne LiDAR point clouds. Instead of designing certain complex and sensitive convergence criteria for iterative plane extraction, we directly determine the instance centers and number of planes included in point clouds, which is the critical prior information of planes, to guide the subsequent segmentation. To achieve that, we propose a dual-branch transformer-based network to simultaneously predict the accurate planar centers and the shifted points distributed around the associated instance centers. Then, we use the robust shifted points near the predicted center to estimate the plane parameters and assign the unlabeled points to the nearest estimated plane to obtain the final roof plane segments. The quantitative and qualitative experimental results on two datasets have shown that our approach can segment roof planes effectively and accurately and achieve the best performance among all the comparison approaches.

Nevertheless, the proposed approach still cannot perform end-to-end extraction of planar patches. Thus, the accumulation of errors at different stages may limit the realization of the optimal segmentation. To achieve further improvement, we aim to develop an end-to-end transformer-based approach for roof plane segmentation in the future work.

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Siyuan You received the B.E. degree from Wuhan University of Remote Sensing Science and Technology, Wuhan, China, in 2024. He is currently pursuing the M.S. degree with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan.

His main research interests include machine learning, light detection and ranging (LiDAR) point clouds processing, instance/semantic segmentation, object detection, and model reconstruction.



Guozheng Xu received the B.E. degree from Sun Yat-sen University, Guangzhou, China, in 2018, and the M.S. degree from Wuhan University, Wuhan, China, in 2021, where he is currently pursuing the Ph.D. degree with the School of Remote Sensing and Information Engineering.

His research focuses on planar optimization, multiview stereo, 3-D modeling, and model optimization, covering computer vision, graphics, and geometric processing.



Pengwei Zhou received the B.S. degree in vehicle engineering from Hubei University of Automotive Technology, Shiyan, China, in 2017, and the M.S. degree from Wuhan University of Technology, Wuhan, China, in 2022. He is currently pursuing the Ph.D. degree with the Computer Vision and Remote Sensing Laboratory, Wuhan University, Wuhan.

His research interests include multisensor spatial–temporal calibration, visual–inertial navigation systems, and structure from motion.



Yubing Wei received the B.E. and M.S. degrees from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2016 and 2019, respectively, where he is currently pursuing the Ph.D. degree.

His main research interests include mesh model processing and simplification.



Jian Yao (Member, IEEE) received the B.Sc. degree in automation from Xiamen University, Xiamen, China, in 1997, the M.Sc. degree in computer science from Wuhan University, Wuhan, China, in 2000, and the Ph.D. degree in electronic engineering from The Chinese University of Hong Kong, Hong Kong, in 2006.

From 2001 to 2002, he has ever worked as a Research Assistant at Shenzhen Research and Development Centre, City University of Hong Kong, Shenzhen, China. From 2006 to 2008, he worked as a Post-Doctoral Fellow with the Computer Vision Group, IDIAP Research Institute, Martigny, Switzerland. From 2009 to 2011, he worked as a Research Grantholder with the Institute for the Protection and Security of the Citizen, European Commission—Joint Research Centre (JRC), Ispra, Italy. From 2011 to 2012, he worked as a Professor with Shenzhen Institutes of Advanced Technology (SIAT), Chinese Academy of Sciences, Shenzhen. Since April 2012, he has been a Hubei "Chutian Scholar" Distinguished Professor with the School of Remote Sensing and Information Engineering and the Director of the Computer Vision and Remote Sensing (CVRS) Laboratory, Wuhan University. He has published over 100 papers in international journals and proceedings of major conferences and is the inventor of over 30 patents. His research interests mainly include computer vision, image processing, deep learning, light detection and ranging (LiDAR) data processing, robotics, and simultaneous localization and mapping (SLAM).



Li Li received the B.E., M.S., and Ph.D. degrees from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2013, 2016, and 2019, respectively.

He is currently an Associate Researcher with the School of Remote Sensing and Information Engineering, Wuhan University. He mainly works on image mosaicking, texture mapping, light detection and ranging (LiDAR) data processing, and machine learning.

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